

Neuro-Symbolic MeMo: Injecting Structured Semantic Representations into Associative Memory Layers for Mechanistic Interpretability

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Abstract

While Transformer-based Large Language Models (LLMs) have achieved remarkable performance, they still operate largely as statistical black boxes. This opacity often leads to uninterpretable reasoning pathways and factual hallucinations. The MeMo architecture recently introduced a compelling alternative: explicitly memorizing sequences of discrete tokens within Correlation Matrix Memories (CMMs) to enable exact recall and localized model editing. However, because MeMo is currently bounded by raw textual strings, it lacks an explicit awareness of structured meaning. In this proposal, I introduce *Neuro-Symbolic MeMo*, a framework that embeds logical predicates and semantic predicate-argument structures from the Uniform Meaning Representation (UMR) framework directly into these associative memory layers. By shifting the model's memorization from surface-level tokens to structured logical facts, generated outputs can be mechanistically traced back to exact logical representations stored in the correlation matrices. This approach offers a theoretically sound pathway for transparent reasoning, factual traceability, and precise machine unlearning.

Keywords

Mechanistic Interpretability, Neuro-symbolic AI, Associative Memory, Uniform Meaning Representation, Machine Unlearning

1. Introduction

Large Language Models (LLMs) continue to advance rapidly, yet their reliance on optimizing billions of continuous parameters leaves them functioning as opaque statistical engines. Standard architectures draw on probability distributions over massive datasets rather than explicit, addressable memory mechanisms. Consequently, they inherently struggle with interpretable reasoning and targeted knowledge editing. Attempting to "unlearn" a specific copyrighted fact or correct an outdated piece of information in a standard Transformer, for example, often risks catastrophic forgetting across the wider network.

Fundamentally rethinking how language models store information offers a way out of this black-box paradigm. Recent proposals advocate for a shift based on the principle that "memorization precedes learning" [1]. By utilizing deterministic outer-product memory matrices rather than distributed continuous weights, it is theoretically possible to guarantee exact structural recall and provide the capacity for precise model editing.

However, existing associative memory models in NLP are primarily text-based, meaning they memorize raw token sequences. To achieve true reasoning capabilities, models must understand the relationships between entities, not just their sequential proximity. To bridge this gap, this paper introduces a novel extension to the associative memory paradigm: *Neuro-Symbolic MeMo*. I propose leveraging the Uniform Meaning Representation (UMR) [2] framework to integrate explicit, structured semantic representations directly into associative memory layers.

By embedding graph-structured logical predicates and semantic relationships into high-dimensional vector spaces, the underlying matrices can memorize structured meaning rather than mere surface-level strings. Drawing on my previous thesis work in cross-domain argumentative structures, this approach also provides a natural mechanism to test whether UMR-grounded memories generalize robustly across varying conversational domains. Ultimately, this project offers a mathematically traceable framework

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where linguistic outputs can be mechanistically audited back to the exact logical structures preserved within memory.

2. Background and Related Work

To contextualize the Neuro-Symbolic MeMo framework, it is necessary to examine current paradigms in both mechanistic interpretability and neuro-symbolic language modeling, highlighting where traditional approaches fall short compared to associative memory mechanisms.

2.1. Post-Hoc vs. By-Design Interpretability

The current dominant paradigm for understanding LLMs involves post-hoc mechanistic interpretability. Researchers attempt to reverse-engineer trained Transformers by identifying induction heads, isolating specific attention layers, or training sparse autoencoders on internal activations to map features to human-understandable concepts. While these efforts yield valuable insights into existing models, they are fundamentally diagnostic—they attempt to explain a black box only after it has been built.

In contrast, the MeMo architecture advocates for interpretability *by-design* [1]. By utilizing Correlation Matrix Memories (CMMs) [3], the storage and retrieval of information are governed by exact linear algebra rather than stochastic gradient descent over opaque hidden layers. My proposed extension builds on this mathematical foundation by ensuring that the linear algebra is operating on interpretable logical structures rather than arbitrary text tokens.

2.2. Neuro-Symbolic Integration Limitations

Current efforts to combine neural language models with symbolic logic largely rely on either Retrieval-Augmented Generation (RAG) or prompting strategies (e.g., Chain-of-Thought). While RAG successfully injects external knowledge (such as knowledge graphs) into the model’s context window, it does not alter the model’s internal parameter weights or its fundamental reasoning mechanics.

Alternatively, some frameworks fine-tune models directly on neuro-symbolic logical representations [4]. However, standard fine-tuning still distributes the knowledge amorphously across the network’s multi-layer perceptrons, making targeted model editing virtually impossible. Neuro-Symbolic MeMo proposes a native integration: compiling the symbolic logic directly into the model’s primary memory matrices, ensuring the logic is intrinsically bound to the architecture.

3. Proposed Methodology: Neuro-Symbolic MeMo

To transition the architecture from surface-level token memorization to structured semantic recall, the proposed research will follow a three-stage pipeline: Graph-to-Vector Embedding, Associative Semantic Storage, and Traceable Retrieval and Unlearning.

3.1. Graph-to-Vector Semantic Embedding

The primary technical challenge in integrating continuous neural networks with discrete associative memories is a fundamental mathematical conflict: standard Graph Neural Networks (GNNs) optimize for semantic similarity, whereas Correlation Matrix Memories (CMMs) require quasi-orthogonality to prevent catastrophic interference. To ensure these two paradigms can coexist, the proposed framework utilizes a strictly sequential, two-step embedding pipeline.

Step 1: Semantic Topology Extraction First, the discrete UMR representation, modeled as a directed graph of logical triples $G = \{(s_i, p_i, o_i)\}_{i=1}^n$, is processed by a GNN. The GNN acts strictly as an intermediate feature extractor. Through message passing, it captures the relational topology of the subject, predicate, and object, outputting a dense, continuous semantic embedding $\mathbf{v}_{dense} \in \mathbb{R}^k$. This step isolates the model’s semantic “understanding” operations.

Step 2: Orthogonal Projection To store this semantic understanding within the MeMo architecture, $\mathbf{v}_{dense}$ must be mapped into a space where independent facts do not mathematically overwrite one another. We apply a static linear projection, multiplying $\mathbf{v}_{dense}$ by a fixed random matrix W_{proj} drawn from a multivariate Gaussian distribution. This operation acts as a Johnson-Lindenstrauss Transform, projecting the dense vector into a high-dimensional space to generate the final query key $\mathbf{q}^{(k)} \in \mathbb{R}^d$. This projection serves as the critical mathematical bridge, adapting the dense semantic output of the GNN into the quasi-orthogonal formatting required for additive storage in the CMM, though this projection alone does not guarantee orthogonality between semantically similar facts (see Section 5).

3.2. Associative Semantic Storage

Once we have generated these semantic embeddings, they are explicitly memorized within the Correlation Matrix Memories. Following standard CMM formalization [3] and aligning with the MeMo key-value conventions, the network stores the association between the semantic query $\mathbf{q}^{(k)}$ and the target outcome $\mathbf{x}^{(k)}$ by updating the correlation matrix M :

$$M = \sum_{k=1}^K c \cdot \mathbf{q}^{(k)} \mathbf{x}^{(k)\top} \quad (1)$$

where K is the total number of semantic facts to be memorized, and c is a normalization scalar. Unlike standard LLMs that distribute knowledge amorphously, this formulation provides a theoretical pathway to localize the logical fact explicitly as an outer product within M . Validating the exactness of this localization under varying memory loads will be a primary empirical objective of this work.

3.3. Mechanistic Traceability and Unlearning

In an optimal, low-interference setting, the output $\mathbf{x}'$ can be mathematically traced back to the specific $\mathbf{q}^{(k)} \mathbf{x}^{(k)\top}$ components that generated a high similarity score. Ideally, this reveals the exact UMR logical graph that triggered the model’s response, though empirical validation against strong baselines is required to measure traceability under heavy memory scaling.

Additionally, this architecture is designed to support targeted machine unlearning [1]. If a specific fact or semantic relationship is deemed incorrect, it can theoretically be erased from the model without retraining. We achieve this by computing the outer product of the obsolete UMR structure and subtracting it from the memory matrix:

$$M_{edited} = M - c \cdot \mathbf{q}^{(obsolete)} \mathbf{x}^{(obsolete)\top} \quad (2)$$

Under the condition that $K \ll d$, the quasi-orthogonality of the embedded space aims to enable the near-complete erasure of the targeted semantic structure. Quantifying the actual preservation of surrounding associative memories during this matrix subtraction remains a key empirical goal of the proposed research.

4. Experimental Plan and Expected Outcomes

Because this proposal outlines a foundational shift toward by-design interpretability, the planned Ph.D. research is structured around three core experimental phases: small-scale validation, baseline benchmarking, and unlearning efficacy.

4.1. Datasets and Baselines

Initial experiments will utilize existing semantic graphs, extracting logical triples from established cross-lingual UMR corpora [2]. The Neuro-Symbolic MeMo will be benchmarked against two primary baselines: (1) the standard text-based MeMo architecture [1], which will help isolate the performance

gain of injecting structured semantics over raw tokens, and (2) standard Transformer models of comparable parameter size, to evaluate exact factual recall against statistical hallucination rates.

4.2. Phase 1: Validation of Scaling Laws

The generalization error bounds predicted by Cabannes et al. [5] apply robustly to standard associative memories; however, whether these bounds transfer automatically to predicate-structured embeddings remains an open theoretical question. Before initiating full-scale training, I plan to build a proof-of-concept pipeline in PyTorch to formally investigate this. The goal is to confirm that the mathematical scaling laws hold true when the input keys $\mathbf{q}^{(k)}$ represent explicitly orthogonalized semantic predicates rather than heavy-tailed text tokens, evaluating memory capacity relative to the embedding dimension d .

4.3. Phase 2: Mechanistic Retrieval Metrics

To properly quantify interpretability, I will evaluate the framework using exact-match accuracy for logical triple recall. Furthermore, I will introduce a traceability metric that measures the cosine similarity between the generated response vector and the exact UMR memory trace stored in the CMM. Ideally, the model should achieve near 100% structural recall for explicitly memorized facts, providing a transparent audit trail for every generated response.

4.4. Phase 3: Zero-Shot Model Editing

Finally, the project will evaluate the machine unlearning mechanism by systematically subtracting targeted semantic triples from the memory matrix M . This matrix subtraction operation should reduce the recall of the targeted fact to zero. To ensure robustness, we will monitor performance on a holdout set of unaffected UMR triples to prove that the matrix subtraction does not induce catastrophic forgetting, thereby proving the viability of zero-shot model editing.

5. Limitations and Future Work

While theoretically robust, integrating continuous GNNs with discrete associative memories introduces significant research challenges that will be the primary focus of future Ph.D. work.

First is the **Orthogonality Paradox**. The Johnson-Lindenstrauss Transform explicitly preserves vector distances. Consequently, if the GNN processes two distinct but semantically similar facts (e.g., "The doctor cured the patient" versus "The physician healed the sick"), the resulting dense vectors will be highly similar. When projected into the MeMo space, they will remain similar rather than quasi-orthogonal, leading to memory interference within the CMM. Future research must explore contrastive learning objectives to force the GNN to produce orthogonal outputs for distinct logical triples.

Second is the challenge of **End-to-End Training Dynamics**. Standard neural networks rely on continuous, differentiable paths for backpropagation. However, the MeMo correlation matrix stores information via discrete, additive outer products. This creates a non-differentiable bottleneck, preventing standard error gradients from passing backwards through the memory matrix to update the GNN's weights. Investigating alternative training mechanisms, such as reinforcement learning or isolated pre-training of the GNN, is necessary to overcome this barrier.

Finally, the reliance on high-quality UMR annotations introduces a data bottleneck, as parsing raw, unstructured text into perfect logical graphs at runtime remains an open challenge in natural language processing.

6. Conclusion

This research proposes *Neuro-Symbolic MeMo* as a foundational extension to the associative memory architecture, transitioning the paradigm of text memorization into the realm of structured logical recall. By injecting UMR graphs directly into Correlation Matrix Memories, this project outlines an algebraically explicit storage mechanism that is natively capable of transparent reasoning, exact factual recall, and zero-shot model editing. This approach aligns seamlessly with the goals of *Mechanistic Interpretability by-design* and *Neuro-symbolic approaches by-design*. Ultimately, this research offers a structurally transparent alternative to the statistical opacity of current language models, ensuring that complex reasoning is both explicit and perfectly auditable.

7. Declaration on Generative AI

During the preparation of this work, the author used generative AI tools to assist with drafting the initial manuscript and structuring the narrative. After using these tools, the author thoroughly reviewed and extensively revised the content to ensure the mathematical and theoretical accuracy of the proposed architecture. The author takes full responsibility for the final content, claims, and original ideas presented herein.

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